	Fig. 3 – Current – Potential Difference Graph
3(a)	The terminal pd across the battery will increase. This is because total resistance of circuit increase causing current to fall. As $V = E - Ir$, potential difference across internal resistance falls causing terminal pd V to increase.
3(b)(i)	$4.0\ \Omega + 3.5\ \Omega + 0.25\ \Omega = 7.75\ \Omega$ current= $\frac{5.0}{7.75} = 0.6452$ A $V = E - Ir = 5.0 - (0.6452 \times 0.25)$ $= 4.8387$ V
3(b)(ii)	Power supplied by battery = $\frac{Emf^2}{R_{tot}} = 3.226$ W Power delivered to ext circuit = $I^2 R_{ext} = (0.6452)^2 (4.0 + 3.5) = 3.122$ W $Efficiency = \frac{P_{EXT}}{P_{EMF}} \times 100 = 96.78\%$

3(c)(i)	$V_{PJ} = \text{pd of } V_{cellC} = E - Ir$ $= E - (0)r$ $= 1.2 \text{ V}$
3(c)(ii)	The potential difference across PQ $V_{PQ} = IR_{PQ} = (0.6452)(3.5) = 2.2582 \text{ V}$ $\frac{V_{PJ}}{V_{PQ}} = \frac{L_{PJ}}{L_{PQ}}$ $L_{PJ} = 0.5314 \text{ m.}$
3(c)(iii)	When J is moved towards Q, the potential difference across PJ increases. As V_{PJ} Becomes greater than the terminal pd of $V_{cell\ C}$, the potentiometer is no longer balanced. Current will flow into the positive terminal of cell C and through the ammeter.
4(a)(i)	The magnetic flux linkage through the coil is $NBA\cos\theta$ where N is the number of